Selective Recovery for Multilingual ASR: Candidate Acquisition and Acceptance

Zhifan Pan[1,2,3], Maihemuti Maimaiti[1,2,3,*], Yingxiong Gao[1,2,3], Yuhui Ming[1,2,3], Aishan Wumaier[1,2,3]
[1]School of Computer Science and Technology, Xinjiang University, Urumqi, China
[2]Xinjiang Multimodal Intelligent Processing and Information Security
Engineering Technology Research Center, Xinjiang University, Urumqi, China
[3]Joint International Research Laboratory of Silk Road Multilingual
Cognitive Computing, Xinjiang University, Urumqi, China
*Corresponding author: mahmutjan@xju.edu.cn

***Abstract*—Fallback recovery depends on both candidate acquisition and acceptance. We isolate these decisions using a text-compression trigger and fixed four-feature selector. On 7,500 frozen Common Voice recordings, selective recovery lowers macro word error rate by 2.26 percentage points and matches unconditional recovery's error counts in all three seeds. Development-exposed CV27 TEST retains 99.98% of pooled positive edit gain. Matched-bank unweighted ranking trails expected risk in every seed; Direct's and edit-weighted ranking's advantages concentrate in individual checkpoints and depend on source. Three-checkpoint stratified timing estimates show up to 5.15% savings over unconditional reranked recovery; Native remains faster. Code: https://github.com/Zhifan-Pan/ISCAS2027.**

***Index Terms*—multilingual speech recognition, selective recovery, candidate acquisition, N-best reranking, confidence estimation, conditional computation**

I. Introduction

Acquisition may preserve corrections under one selector but discard them under another. We separate acquisition from acceptance to study this dependence.

Multilingual N-best reranking [1], [2], discriminative rescoring [3]–[5], voting [6] and confusion networks [7] exploit existing hypotheses. Generative correction expands banks [8]–[11]; risk-based pruning and ASR sampling allocate computation [12], [13].

Conditional computation uses recognizer cascades [14], selective passes [15], encoder proxies [16], selective correction [17] or language-specific distillation [18]. Confidence need not predict computational benefit [19]: acceptance determines which acquired hypotheses improve output.

With recognizer and language labels frozen, we compare acquisition within sources. A frozen panel evaluates the original policy; retrospective controls assess acceptance objectives [20]–[22] and acquisition budgets.

II. Controlled Acquisition and Acceptance

A. Recording-level acquisition

Normalizing and deduplicating five beams per window i gives B_i, retaining first occurrences and sources; b_i is the raw top beam. Fallback g_i gives $F_i = B_i \cup \{g_i\}$, at most six candidates. The outer $\mathrm{CR}_{\mathrm{text}}$ divides top-beam UTF-8 byte length by zlib-compressed length, or is zero for empty bytes. With $z_i = \mathbf{1}\{\mathrm{CR}_{\mathrm{text},i} > 2.4\}$ and $z^{\mathrm{rec}} = \max_i z_i$, any trigger acquires fallback for every window, including singletons.

Recovery uses final Hugging Face fallback at temperatures $0, 0.2, \ldots, 1$, with compression, log-probability and no-speech thresholds 1.35, -1 and 0.6. Internal compression uses serialized token-ID bytes, unlike the outer text scale. Transformers v4.57.6 lists 1.35 as common [23]. Native HF uses the same configuration. Retries are conditional; intermediate hypotheses never enter the bank.

B. Four-feature Shared acceptance

For $f(y) = (a, p, t, r)$, decoder score a averages teacher-forced log probabilities over T body tokens, excluding prefix, padding and end tokens; $t = \log(1+T)$. XLM-R-base pseudo-log-likelihood p [24], [25], or PLL, averages log probabilities of individually masked nonspecial subwords in the original string, using centered windows of at most 510 subwords rather than within-word masking [26]. No body tokens gives $a = t = 0$; no PLL targets gives $p = 0$. Normalization-only emptiness need not give zero features.

For normalized words $w_{1:M}$, define the overlapping n-gram count $c_y(u) = \sum_{k=1}^{M-n+1} \mathbf{1}\{w_{k:k+n-1} = u\}$. Repetition is

$$r_{n,h}(y) = \frac{\sum_{j=1}^{M-n+1} \mathbf{1}\{c_y(w_{j:j+n-1}) \geq h\}}{\max(1, M-n+1)}. \tag{1}$$

We use $r = r_{4,3}$, which is zero for $M < 4$. Repetition is also a target of unlikelihood training [27].

Centering $x_i(y) = f(y) - f(b_i)$ preserves rankings and softmax probabilities. Feature-wise RMS scales $s \in \mathbb{R}^4_{>0}$ use training-fold pairs $\{(i,k) : i \in \bigcup_{\ell=1}^{L} \mathcal{I}_\ell,\ y_{ik} \in C_i \setminus \{b_i\}\}$ in fitting banks C_i; scales below 10^{-12} become one. With edit benefit $\Delta_{ik} = E(b_i, y_i^*) - E(y_{ik}, y_i^*)$ and $\pi_{ik} = \mathrm{softmax}_k\big(\theta^\top \mathrm{diag}(s)^{-1} x_{ik}\big)$, expected-risk fitting [20] minimizes

$$\mathcal{L}(\theta) = -\frac{1}{L} \sum_{\ell=1}^{L} \frac{\sum_{i \in \mathcal{I}_\ell} \sum_k \pi_{ik} \Delta_{ik}}{N_\ell} + \lambda \|\theta\|_2^2. \tag{2}$$

Here $\mathcal{I}_\ell = \{i : \mathrm{lang}(i) = \ell\}$ indexes fitting windows, N_ℓ counts reference words and $L = 5$.

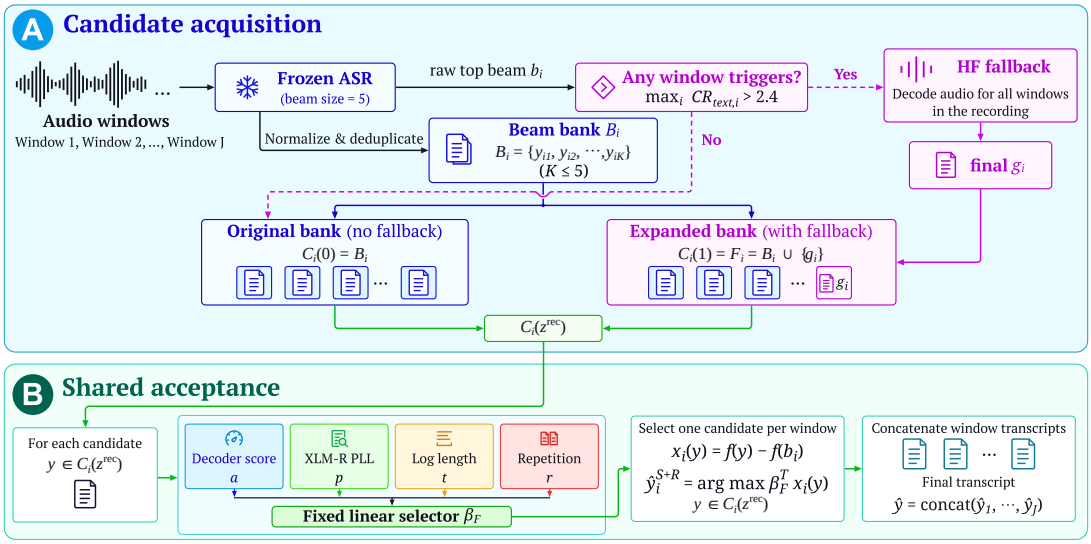


Fig. 1. Acquisition (A) expands every window's candidate bank if any window triggers. Acceptance (B) applies the same fixed four-feature selector to the original or expanded bank. Selected window transcripts are then concatenated.

Shared weights $\beta_F = \text{diag}(s)^{-1}\theta$ are fitted per run on greedy-expanded DEV and fixed across sources, with bank-order ties. Let $\hat{y}_i^B = \arg\max_{y \in B_i} \beta_F^\top x_i(y)$, $C_i(0) = B_i$ and $C_i(1) = F_i$. Then

$$\hat{y}_i^{S+R} = \arg \max_{y \in C_i(z^{\text{rec}})} \beta_F^\top x_i(y),$$
$$\hat{y}_i^{\text{Direct}} = \begin{cases} \hat{y}_i^B, & z^{\text{rec}} = 0, \\ g_i, & z^{\text{rec}} = 1. \end{cases} \tag{3}$$

B reranks B_i; N+R requests and reranks F_i unconditionally. Direct bypasses selection and PLL on triggers, retaining B elsewhere. CV27 has $z^{\text{rec}} = z_i$; multi-window FLEURS uses the recording flag. Concatenated transcripts receive complete-reference scoring. Shared fixes acceptance to isolate acquisition.

C. Matched acceptance controls

G4 is the original β_F. H4 refits expected risk on complete HF-expanded DEV banks; H3 uses the same objective with (a, t, r). HR4 fits RankNet [21] on H4's banks and four features. Its logistic loss favors lower-edit candidates in unequal-edit pairs, with language-reference normalization and equal total bank weight. HWR4 additionally weights pairs by $|E_j - E_k|/c$, where c is the training-fold mean edit gap under the original pair weights.

Five component folds group clients and reference texts within languages. RMS uses training folds; held-fold hard macro WER chooses among 49 log-spaced penalties from 10^{-4} to 1, favoring stronger regularization on ties. Full-DEV fitting uses zero-initialized L-BFGS-B. Singletons retain errors and reference denominators but contribute no ranking pairs.

Same-bank minimum Bayes risk (MBR) [22] minimizes expected word edits. U-MBR is uniform; D-MBR uses body scores $\text{softmax}(Ta/\tau)$, not calibrated sequence posteriors. DEV WER selects $\tau \in \{0.1, 0.3, 1, 3, 10\}$, favoring larger values on ties. Candidate ties retain bank order.

A *complete policy* uses its acceptor on both banks. A *triggered-subset evaluation* changes acceptance only on triggers, retaining G4 elsewhere and full-panel denominators. H3Direct uses H3 off-trigger and final HF on-trigger.

III. Data and Evaluation

We adapt Whisper-small [29] on 44.60 h of FLEURS TRAIN [28] to Arabic, Persian, Polish, Turkish and Italian using LoRA [30] with three different random seeds: 4,656 updates each, rank and scale four, zero dropout, all attention projections. Adaptation excludes CV27 TRAIN; reported fits exclude FLEURS DEV. Selectors and gates use CV27 DEV. Complete TEST has 3,665 FLEURS and 58,514 Common Voice 27.0 recordings [31] from release `2026-09-11`. TEST informed development, making comparisons retrospective.

Confirmation uses 7,500 CV27 TRAIN recordings, 1,500 per language, excluding previously accessed recordings, audio and reference texts. Predictions, stopping rules and the 0.10 pp one-sided noninferiority margin were locked before reference evaluation under procedural controls. Within-panel correlation is addressed by 10,000 synchronized within-language speaker-component bootstrap draws. Recurring Common Voice speakers across splits restrict generalization to the observed speaker-clustered sampling frame. Later gates, refits and window policies remain retrospective.

Decoding uses known languages, bfloat16, five beams, length penalty one, native suppression and a 444-token cap in Transformers 4.57.6. At 16 kHz, N samples yield $J = \lceil N/480000 \rceil$ windows bounded by $\lfloor jN/J \rfloor$, without previous-window text conditioning. Deduplication, WER and r share NFC normalization, digit unification, punctuation-to-space, Persian mapping and Turkish casing [32]. Source RNG mappings stay fixed.

WER aggregates within languages before averaging five languages and three seeds; calls pool seed–recordings. Seed diagnostics use consistent order. TEST intervals use 2,000 paired language-internal client/reference-component draws, synchronized across seeds and conditional on frozen fits and source draws. FLEURS controls are descriptive because component metadata are unverified.

Costs use 255 FLEURS and 472 CV27 recordings: all first-checkpoint triggers or Native-output changes plus ten ordinary recordings per language–duration-quintile cell, weighted by inclusion probability. Unsampled later-checkpoint triggers number five on FLEURS and 207 on CV27, or 0.05% and 0.12% of seed–recordings in probability-sampled strata. Triggered N+R and S+R perform identical acquisition and scoring operations. Both studies use three randomized paired A40 rounds, resident models, encoder/first-search reuse and 32-mask PLL batches. Synchronized waveform-to-output timing excludes loading, warmup and interfered blocks. Session-conditional intervals use 4,000 paired stratified bootstrap draws with finite-population correction and equal checkpoint weights synchronized in A.

IV. Results

A. Acquisition with fixed acceptance

TABLE I
WER (%); PP GAINS AND CONTRASTS COMPUTED BEFORE ROUNDING. CHECKS: HF ACQUISITION, CR GATE, R RESELECTION; †: R OFF-TRIGGER.

Path	HF	CR	R	Scorer	FLEURS	CV27
Raw beam-5 top-1	–	–	–	–	18.90	36.41
Native HF	✓	–	–	–	18.55	33.80
B	–	–	✓	G4	17.45	33.98
N+R	✓	–	✓	G4	17.09	31.68
S+R	✓	✓	✓	G4	17.09	31.68
Direct	✓	✓	✓†	G4	17.09	31.46
H3Direct	✓	✓	✓†	H3	18.68	33.64
Reference-assisted lower bounds on TEST						
Original-bank oracle					16.19	30.19
Full-HF oracle					15.83	27.68
CR-routed oracle					15.83	27.68
S+R gap to routed oracle, pp					1.26	4.00
Frozen CV27 confirmation: 7,500 recordings						
					B	N+R = S+R	Gain
Macro-average					35.02	32.75	2.26
Contrast						ΔWER	95% upper bound
S+R − B						-2.26	-1.32
S+R − N+R						0.00	0.00

Frozen-panel S+R gains 2.26 pp over B: 2.04, 0.20 and 4.54 pp by seed. Substitution, deletion and insertion counts match N+R in every seed, meeting the prespecified 0.10 pp margin. Among 61 triggers, 49 improve and two worsen edits versus B. The zero-loss interval remains panel-specific.

Post-lock Direct has 32.71% WER and S+R 32.75%; both beat B and Native's 34.87% by over 2.1 pp. From unrounded means, Direct minus S+R is -0.05 pp, with retrospective paired speaker-bootstrap 95% CI $[-0.19, +0.04]$.

TABLE II
SAME-HF TEST WER (%); GAIN B−S (PP) BEFORE ROUNDING. CHECKS: PLL, PAIRWISE P, EDIT-GAP WEIGHTING W.

				FLEURS			CV27		
Acceptor	PLL	P	W	B	S	Gain	B	S	Gain
G4	✓	–	–	17.45	17.09	0.37	33.98	31.68	2.30
H4	✓	–	–	17.46	17.09	0.37	33.97	31.52	2.45
HR4	✓	✓	–	17.51	17.14	0.37	34.02	32.30	1.72
H3	–	–	–	19.05	18.68	0.37	36.15	33.77	2.38
HWR4	✓	✓	✓	17.47	17.10	0.37	33.97	31.44	2.53
U-MBR	–	–	–	19.14	19.14	0.00	36.91	36.88	0.03
D-MBR	–	–	–	18.92	18.75	0.17	36.09	34.48	1.60

CV27 ΔWER [individual 97.5% CI]: original family	
Triggered HR4 − H4	+0.72 [+0.40, +1.16]
Complete H3 − H4	+2.25 [+2.16, +2.34]
Later two-comparison family	
Triggered U-MBR − H4	+2.51 [+1.93, +3.18]
Triggered HWR4 − H4	-0.08 [-0.15, -0.03]

The frozen archive locked G4 scores and transcripts to isolate acquisition; refits use full TEST.

TEST Shared retains 99.98% of CV27 positive edit gain with full-language denominators. FLEURS retains all 10 beneficial instances at 0.09% calls; CV27 retains 370 of 373 at 0.30%, missing five edits. N+R and S+R outputs differ after normalization on one FLEURS and 13 CV27 nontriggers despite nearly identical WER.

Reference-assisted oracles minimize full-reference edits over original, expanded or CR-routed banks, including concatenated two-window candidates. Routing changes the bound by under 0.01 pp; CV27 S+R remains 4.00 pp above it. Top-1 to B measures reselection, not PLL alone.

B. Acceptance changes recovery opportunities

Bypassing the greedy-trained selector on triggers, Direct lowers CV27 WER while retaining 67.13% of its own positive edit gain, against Shared's 99.98%. Relative to unconditional final HF, it forgoes 1.24 pp of beneficial changes but avoids 3.58 pp of harmful changes.

Shared beats Direct by 0.53 pp for greedy recovery and 0.43 pp for one $T = 0.2$ draw. HF reverses this ranking by 0.22 pp, with 95% CI [0.11, 0.37]. Direct-minus-Shared differences of -0.68, $+0.00$ and $+0.02$ pp by seed concentrate the advantage in one checkpoint. DEV favored HF reselection at 30.08% versus 30.09%, motivating fixed acceptance.

C. Acceptance on matched HF banks

With the same four-feature architecture, HWR4 achieves the lowest complete CV27 WER, 31.44%, against Direct's 31.46%. Triggered HR4 minus H4 is positive in every seed: $+0.91$, $+0.08$ and $+1.17$ pp. Edit-gap weighting reverses the mean to -0.08 pp, concentrated in one checkpoint: 0.00, -0.25 and $+0.01$ pp for HWR4 minus H4.

FLEURS gains round to 0.37 pp for all five acceptors: off-trigger S equals its own B; on one of ten triggers, H3's B has two fewer errors while S error counts match. Language and checkpoint averaging reduces this gain difference below 0.006 pp.

TABLE III
POPULATION-WEIGHTED MEAN MS; OBSERVED PEAK ALLOCATED MIB WITH BOTH MODELS RESIDENT. THREE PAIRED ROUNDS PER STUDY.

	FLEURS		CV27	
Path	ms	MiB	ms	MiB
A: Four paths, three checkpoints				
Native HF	1291.7	2980.8	517.4	2813.8
B	1352.7	3144.9	591.9	3133.1
N+R	1436.8	2988.3	639.2	3133.1
S+R	1362.8	3189.7	610.0	3133.1
B: Seven paths, one checkpoint				
Native HF	1304.8	2978.4	531.2	2811.8
Raw top-1	1222.6	2946.6	479.9	2571.0
B	1364.0	3142.4	605.1	3131.1
N+R	1449.6	2986.3	657.9	3131.1
S+R	1374.5	3187.2	628.0	3131.1
Direct	1372.8	3187.2	625.8	2811.8
H3Direct	1314.0	3187.2	586.4	2811.8
S+R saving versus N+R: % [pointwise 95% CI]				
A	5.15 [2.84, 9.41]		4.57 [4.40, 4.73]	
B	5.18 [2.88, 9.39]		4.54 [4.44, 4.66]	
Within-study B strategy differences, not stage times				
B − Raw top-1, ms	141.4		125.2	
N+R − B, ms	85.6		52.8	
S+R − B, ms	10.5		22.9	
Increment saved	87.7%		56.6%	

Table II gives 95% Bonferroni coverage per family; the later family uses 10,000 synchronized draws.

The MBR controls also trail H4 [13]. H3-based policies trail H4 and Direct despite baseline gains. Nontriggers account for 97.23% of the refitted H3–H4 gap, measuring neither PLL causality nor time share.

D. Cost, heterogeneity and supporting diagnostics

For systems using B, S+R preserves N+R's accuracy gain while reducing recovery overhead; Native and B remain faster. Study B's weighted B-minus-Raw increments are 141.4 ms on FLEURS and 125.2 ms on CV27; S+R saves 87.7% and 56.6% of their N+R increments over B (Table III). In 25 unweighted CV27 recordings, teacher scoring averages 85.9 ms and PLL 149.3 ms; two singletons skip both. These profile means describe a subset; strategy increments weight all strata.

Study B's Raw top-1 and H3Direct skip PLL. H3Direct saves 6.29% versus Direct but trails Native by 55.2 ms on CV27, with 95% CI [52.3, 58.2]; FLEURS' interval spans zero. H4 and HWR4 change G4's coefficients within the same four-feature scoring architecture; Table III measures the listed paths. Trigger-selective PLL remains a future optimization.

Empty TEST candidates enter only through HF fallback, with $f(\emptyset) = (0,0,0,0)$ and centered score $\beta_F^\top x_i(\emptyset) = -\beta_F^\top f(b_i)$. The margin $-\beta_F^\top f(y) > 0$ holds under signs $(+,+,-,-)$ when $a, p \le 0$, $t > 0$ and $r \ge 0$, even at $r = 0$. Centering preserves this preference, not a transcript probability.

Reverting all 18 empty HF acceptances in 11 CV27 recordings to B gives a deterministic nonempty guard. It retains 91.52% of CV27 gain: 2.10 pp at 31.88% WER.

TABLE IV
TEST GAIN: SEED MEAN; CALLS: POOLED SEED–RECORDINGS.

	FLEURS		CV27	
Language	Gain, pp	Calls, %	Gain, pp	Calls, %
Arabic	0.64	0.23	1.40	0.11
Persian	1.20	0.27	9.05	1.15
Polish	0.00	0.00	0.30	0.33
Turkish	0.00	0.00	0.51	0.03
Italian	0.00	0.00	0.23	0.02
Empty-HF acceptance: 18 instances, 11 CV27 recordings				
Acceptance minus B reversion	$S:-118$		$D:+120$	$I:-1995$

The insertion–deletion tradeoff (Table IV) is consistent with hallucination suppression [33]; acoustic interpretation requires silence labels.

H4+e refits with $e(y) = \mathbf{1}\{y = \emptyset\}$ on H4's DEV folds and grid. With one or two DEV empties per checkpoint, it matches H4's rounded 31.52% CV27 WER and retains all 18 empty acceptances.

Persian supplies 78.77% of CV27 TEST gain, with 9.05 pp at 1.15% calls, and 84.07% of confirmation gain. Polish, Turkish and Italian have zero FLEURS calls and gains. Script, noise and LoRA effects remain coupled (Table IV).

FLEURS' 22 two-window recordings yield 52.69% of net gain. Cached full-TEST local Shared preserves counts, cutting requests from 15 to 10: four to two, three to two and eight to six by seed. Local Direct changes WER by under 0.01 pp; live RNG may differ.

Bulgarian and Croatian greedy WER stays at 37.51%. The earlier 4,431-recording panel gives S+R-minus-B of -0.08 pp, with 95% CI $[-0.23, 0.00]$. SeamlessM4T [34], [35] greedy CV27 WER falls from 12.96% to 12.69% at 0.08% calls. DEV-tuned MBR reaches 12.55% against risk's 12.66% on greedy-expanded banks; full HF remains unvalidated.

HF12 ridge regression predicts fixed-G4 edit gain on grouped CV27 DEV folds from twelve pre-acquisition features: $\log(1+d)$ for duration d in seconds, $\log(1+|B_i|)$, top-two decoder-score gap, softmax-a entropy, population standard deviation and range of a, mean $|a - a(b_i)|$, ranges of t and r, top-beam $\mathrm{CR}_{\mathrm{text}}$ and $r(b_i)$, and maximum repetition. Conditional on G4's fit, predicted negative gain below DEV's $q = 3\times10^{-3}$ quantile triggers multi-candidate windows. HF12 and CR yield 31.71% WER at 0.29% and 0.26% CV27 calls. Their interval spans zero without establishing equivalence; equal-call ranking favors HF12. FLEURS calls are 178, two and five by seed.

V. CONCLUSION

Sparse acquisition matches unconditional recovery across three frozen-panel seeds and reduces reranked-pipeline latency. Expected risk consistently beats unweighted ranking; Direct and edit-weighted ranking depend on checkpoint and source.

ACKNOWLEDGMENT

Supported by the National Natural Science Foundation of China under Grant 62466058 and Autonomous Region Key R&D Program under Grant 2025B03042-1.

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